

General Equilibrium Trade Policy Analysis with Structural Gravity

A fully derived study companion to Chapter 2 of the UNCTAD–WTO
Advanced Guide to Trade Policy Analysis: The Structural Gravity Model
(Yotov, Piermartini, Monteiro and Larch, 2016)
with lesson quizzes and a final exam

Study notes

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Part I

The General Equilibrium Structure of Gravity

Chapter 1 built the structural gravity system and estimated its *partial*, bilateral effects. But a change in trade costs between any two countries in the world ripples through *every other* country's trade, via the multilateral resistance terms. This Part opens up that ripple: what the multilateral resistances mean, how a trade-policy shock propagates through the system in three distinct channels, and how the whole apparatus nests inside richer, dynamic general-equilibrium models.

1 Lesson 1: The general equilibrium gravity system and six properties of the multilateral resistances

Definition: The full structural gravity system

Carrying forward the notation of Chapter 1, with N countries, iceberg trade costs t_{ij} , elasticity of substitution σ , and CES taste parameters α_i :

$$X_{ij} = \frac{Y_i E_j}{Y} \left(\frac{t_{ij}}{\Pi_i P_j} \right)^{1-\sigma} \quad (2-1)$$

$$\Pi_i^{1-\sigma} = \sum_j \left(\frac{t_{ij}}{P_j} \right)^{1-\sigma} \frac{E_j}{Y} \quad P_j^{1-\sigma} = \sum_i \left(\frac{t_{ij}}{\Pi_i} \right)^{1-\sigma} \frac{Y_i}{Y} \quad (2-2, 2-3)$$

$$p_i = \left(\frac{Y_i}{Y} \right)^{\frac{1}{1-\sigma}} \frac{1}{\alpha_i \Pi_i} \quad E_i = \phi_i Y_i = \phi_i p_i Q_i. \quad (2-4, 2-5)$$

Equations (2-4)–(2-5) are new relative to Chapter 1: (2-4) is the market-clearing factory-gate price, and (2-5) links expenditure to nominal income through the exogenous deficit/surplus parameter ϕ_i and *fixed* endowment Q_i (an *endowment economy*, as in Anderson, 1979, and Anderson and van Wincoop, 2003).

Intuition: Six reasons the multilateral resistances are the heart of this chapter

1. **Intuitive trade-cost terms:** all else equal, two *remote* countries (high Π_i, P_j) trade *more* with each other, because they have fewer good alternatives — Krugman's (1995) “two economies on Mars” thought experiment.
2. **Theory-consistent aggregates:** Π_i, P_j collapse the $N \times N$ web of bilateral costs into a $2 \times N$ system of country indexes.
3. **General equilibrium trade-cost terms:** a bilateral liberalization between i and j (i) directly affects i, j ; (ii) ripples to *every* other country through Π_n, P_n ; (iii) potentially feeds back onto i, j themselves.
4. **Decompose incidence:** Π_i is the incidence of trade costs *on producers* in i (as if selling to one world market); P_j is the incidence *on consumers* in j (as if buying from one world market).
5. **Straightforward to construct:** given data on Y_i, E_j and σ , systems (2-2)–(2-3) can be solved by iteration, as a quadratic system in the power transforms $\Pi_i^{1-\sigma}, P_j^{1-\sigma}$, or

— most conveniently — read directly off PPML fixed effects (Lesson 7).

6. **Practically appealing:** they summarize and decompose the incidence of trade policy for policy analysis, and (via Frankel–Romer-style structural equations) can even help identify σ itself.

Derivation: Normalisation: why one degree of freedom is always missing

System (2-2)–(2-3) is homogeneous: if $\{\Pi_i^0, P_j^0\}$ solves it, so does $\{\lambda \Pi_i^0, P_j^0/\lambda\}$ for any $\lambda > 0$ (Anderson and Yotov, 2010b). A normalisation is required — typically $P_R = 1$ for a reference country R , chosen either for data reliability or because it is expected to be little affected by the counterfactual shock (so relative changes approximate absolute changes). **All multilateral-resistance-based indexes are therefore only interpretable relative to the reference country.**

Understanding Quiz 1 — Lesson 1

- Q1.1** All else equal, two very *remote* countries (high Π_i and P_j) will trade with each other:
- (A) less, because remoteness always deters trade
 - (B) more, because they have fewer good alternative trading partners
 - (C) the same as any other pair
 - (D) never, by definition
- Q1.2** The multilateral resistance system (2-2)–(2-3) can only be solved:
- (A) in absolute levels directly, with no ambiguity
 - (B) up to a normalising scalar λ , requiring a reference-country normalisation
 - (C) if $\sigma = 1$
 - (D) using cross-sectional data only
- Q1.3** Π_i is best described as the incidence of trade costs on:
- (A) consumers in country i , as if buying from one world market
 - (B) producers in country i , as if selling to one world market
 - (C) government revenue in country i
 - (D) foreign consumers only

2 Lesson 2: Three channels — partial, conditional general, and full endowment general equilibrium

Intuition: A hypothetical RTA between i and j , three ways

Consider a bilateral trade-cost cut ($t_{ij} \downarrow$) between countries i and j — an RTA. The structural gravity system (2-1)–(2-5) decomposes its total impact into **three channels**, of increasing scope and increasing computational demand.

2.1 Channel 1: Partial equilibrium (direct) effects

Derivation: Direct trade creation

Holding Y_i, E_j, Y, Π_i, P_j all fixed, equation (2-1) alone implies: a 10% fall in t_{ij} raises bilateral trade X_{ij} by $(0.9^{1-\sigma} - 1) \times 100\%$. **This is the entire effect for a naive, atheoretical gravity user** — and it is, by construction, zero for every country other than i and j .

2.2 Channel 2: Conditional general equilibrium effects

Intuition: Why “conditional”?

This scenario holds Y_i, E_j (output, expenditure) fixed — hence “conditional” — but lets the change in t_{ij} propagate through *all* the multilateral resistances $\{\Pi_n, P_n\}_{n=1}^N$ via equations (2-2)–(2-3). It is a genuinely general-equilibrium object even though sizes are frozen, because it captures how the trade-cost cut reshapes *everyone’s* market access.

Derivation: First-order effects on the liberalizing (member) countries

Both Π_i and P_j **fall** for the liberalizing pair (their own trade got cheaper, on average). But equation (2-1) shows this pulls in the *opposite* direction of the direct effect: a lower P_j means country j imports *less* from every source, including i — itself now relatively less special. This is **trade diversion away from third countries, towards i** . By construction, however, $|\Delta t_{ij}| > |\Delta \Pi_i|, |\Delta P_j|$ always, since each multilateral resistance is a size-weighted average across *all* of a country’s bilateral costs, of which the i – j link is only one term. Trade diversion therefore never overturns the direct trade-creation effect.

Derivation: Second-order effects on non-member countries

Non-members’ Π_n, P_n typically *rise* (they become relatively more remote as i, j integrate). This raises trade *among* non-members (trade creation among outsiders) but lowers trade *between* members and non-members (trade diversion away from outsiders). Because part of the diverted trade is absorbed by increased *domestic* sales in non-member countries, the net effect on a non-member’s *international* trade is usually negative — though in principle a non-member with negligible initial trade with the liberalizing bloc could see a small positive net effect via redirected trade with other outsiders. Changes in non-members’ multilateral resistances further feed back onto the members — “third-order” effects, always dominated in magnitude by the first- and second-order effects above.

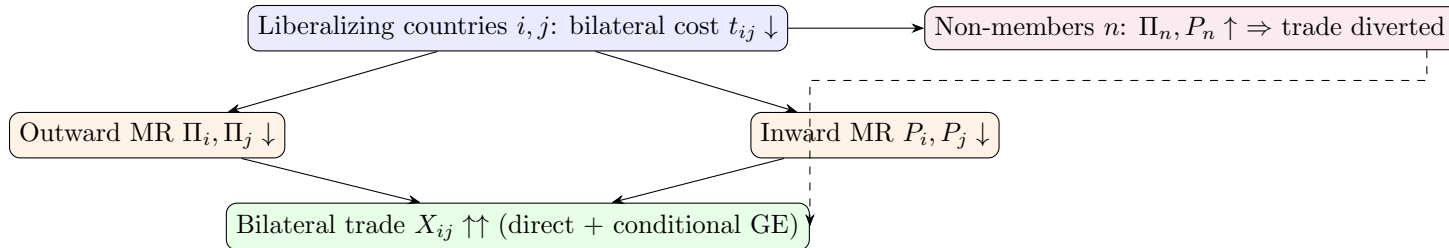


Figure 1: Conditional general equilibrium: a bilateral shock ripples through every country’s multilateral resistances.

2.3 Channel 3: Full endowment general equilibrium effects

Intuition: Why “full endowment”?

This channel additionally **endogenizes** factory-gate prices p_i (via 2-4) and hence nominal output/expenditure Y_i, E_i (via 2-5), while still holding the physical *endowment* Q_i fixed — hence “full endowment” as opposed to a fully endogenous-production model (Lesson 3).

Derivation: The chain of full endowment effects

1. A fall in Π_i (member’s outward resistance) raises i ’s factory-gate price p_i via (2-4): producers internalize their improved market access by charging more.
2. Higher p_i raises $Y_i = p_i Q_i$ and, via (2-5), E_i : members become *effectively richer*. Non-members’ Π_n typically rises, so their p_n, Y_n, E_n *fall* — non-members become effectively poorer.
3. These changes in Y_i, E_j feed back directly into trade via (2-1): richer members trade more with *everyone*, not just each other, while poorer non-members trade less with everyone. Because the size effect on members dominates the size effect on non-members, trade between members and non-members can rise on this channel even though it fell on the “conditional” (trade-diversion) channel.
4. The changes in Y_i, E_j feed back *again* into the multilateral resistances (2-2)–(2-3), since larger countries get more weight in everyone’s market-access aggregate — a third round of (smaller) adjustment.

Intuition: The escalating scope of the three channels

Partial equilibrium: *only i, j* affected. Conditional GE: *every* country’s trade costs affected, sizes frozen. Full endowment GE: *every* country’s trade costs *and* sizes affected, feeding back on each other until the system settles. Each channel is a strict superset of the information in the previous one — which is exactly why Chapter 1’s partial-equilibrium estimates, however carefully estimated, cannot by themselves answer “what happens to world trade and welfare?”

Understanding Quiz 2 — Lesson 2

Q2.1 In the “conditional general equilibrium” scenario, which variables are held fixed?

- (A) bilateral trade costs t_{ij}
- (B) output Y_i and expenditure E_j
- (C) the multilateral resistances
- (D) nothing is held fixed

Q2.2 Trade diversion in the conditional GE scenario can never overturn the direct trade-creation effect between the liberalizing partners because:

- (A) trade diversion does not exist
- (B) the change in each multilateral resistance is a size-weighted average across *all* bilateral costs, so it is always smaller than the direct bilateral change

- (C) σ is always equal to one
- (D) non-members do not respond

Q2.3 In the full endowment GE scenario, a fall in a liberalizing country's outward multilateral resistance Π_i leads to:

- (A) a fall in its factory-gate price and output
- (B) a rise in its factory-gate price, and hence in its nominal output/income
- (C) no change in prices
- (D) a fall in world output

Q2.4 The scope of the three general equilibrium channels, from narrowest to broadest, is:

- (A) full endowment GE, conditional GE, partial equilibrium
- (B) partial equilibrium, conditional GE, full endowment GE
- (C) conditional GE, full endowment GE, partial equilibrium
- (D) they all have identical scope

3 Lesson 3: Nesting gravity inside a dynamic model of growth

Intuition: Beyond the endowment economy

Everything so far treats each country's physical output Q_i as fixed. Anderson, Larch and Yotov (2015c) show the *same* structural gravity system can be nested — with no modification — inside a dynamic Lucas–Prescott-style model where households choose both consumption *and investment*, and capital accumulates over time. This matters because it lets trade liberalization affect growth, not just the static allocation of a fixed pie.

Derivation: The dynamic gravity system

Representative households in j maximize $\sum_{t=0}^{\infty} \gamma^t \ln(C_{j,t})$ subject to $Y_{j,t} = P_{j,t}C_{j,t} + P_{j,t}\Omega_{j,t}$, where $C_{j,t}$ and $\Omega_{j,t}$ (consumption and investment) are each CES aggregates over source-country varieties, exactly as in the static model. Production now uses both labour and capital, $Y_{j,t} = p_{j,t}A_{j,t}L_{j,t}^{1-\eta}K_{j,t}^{\eta}$, and capital accumulates as $K_{j,t+1} = \Omega_{j,t}^{\delta}K_{j,t}^{1-\delta}$. Solving this problem delivers a dynamic gravity system whose **first four equations are identical to the static system** (2-1)–(2-4), plus:

$$Y_{i,t} = p_{i,t}A_{i,t}L_{i,t}^{1-\eta}K_{i,t}^{\eta}, \quad E_{i,t} = \phi_i Y_{i,t}, \quad K_{i,t+1} = \gamma \delta \phi_{i,t} \left[\frac{\eta p_{i,t} A_{i,t} L_{i,t}^{1-\eta} K_{i,t}^{\eta-1}}{(1 - \gamma + \delta \gamma) P_{i,t}} \right] K_{i,t}.$$

Setting $\delta = 0$ (no capital accumulation) collapses this exactly back to the static endowment system.

Intuition: Two new channels: prices feed growth, growth feeds trade

The capital accumulation equation reveals **two** new links. *First*, capital accumulation is **increasing in the factory-gate price** $p_{i,t}$: a higher price raises the marginal product of capital, stimulating investment — so a trade-cost shock *anywhere* in the world, by moving factory-gate prices everywhere (Lesson 2), changes investment everywhere too. *Second*,

capital accumulation is **decreasing in the inward multilateral resistance** $P_{i,t}$: since $P_{i,t}$ prices *both* consumption *and* investment goods, expensive investment goods directly discourage capital formation, while expensive consumption goods raise the opportunity cost of saving today (crowding out investment in favour of current consumption). The resulting capital changes then feed back into $Y_{i,t}$ (via the production function) and hence into trade (2-1) and the multilateral resistances (2-2)–(2-3) — new *direct* and *indirect* size effects, layered on top of the static full endowment GE channels of Lesson 2. In particular, an RTA can raise investment (and hence long-run trade and income) even in *non-member* countries, to the extent that cheaper consumption there frees resources for saving — though this is offset by tougher competition from more efficient member-country producers.

Understanding Quiz 3 — Lesson 3

Q3.1 Setting the capital-share/depreciation parameter $\delta = 0$ in the dynamic gravity system:

- (A) breaks the model
- (B) collapses it exactly back to the static endowment-economy gravity system
- (C) eliminates trade entirely
- (D) only affects consumption, not trade

Q3.2 Capital accumulation in country i is increasing in the factory-gate price $p_{i,t}$ because:

- (A) higher prices lower the marginal product of capital
- (B) higher prices raise the marginal product of capital, stimulating investment
- (C) prices do not affect capital accumulation
- (D) it is a pure accounting identity

Q3.3 Capital accumulation is decreasing in the inward multilateral resistance $P_{i,t}$ because $P_{i,t}$ prices:

- (A) government spending only
- (B) both consumption and investment goods, so expensive $P_{i,t}$ discourages saving/investment
- (C) labour only
- (D) nothing relevant to households

Part II

General Equilibrium Indexes

Once a trade-policy shock is fed through the system of Part I, the researcher needs summary numbers to report. This Part catalogs the standard general equilibrium indexes, split into those that require the normalisation of Lesson 1 and those that are, by construction, normalisation-free.

4 Lesson 4: Indexes that require normalisation

Definition: Inward and outward multilateral resistance indexes

P_j , re-derived as a CES price aggregator $P_j = (\sum_i (\alpha_i p_i t_{ij})^{1-\sigma})^{1/(1-\sigma)}$, can be read as a **supplier access index** (Redding and Venables, 2004): it summarizes the incidence of trade costs on j 's *consumers*. In principle it should track a consumer price index (CPI), but in practice it need not track closely, because (i) it also prices intermediate-goods use invisible in headline CPI, (ii) it is production-weighted rather than final-basket-weighted, (iii) it can absorb home-bias in preferences, and (iv) the underlying CES structure may be mis-specified. Symmetrically, Π_i is a **market access index** for *producers* (Redding and Venables, 2004) or a total-factor-productivity friction in distribution (Anderson and Yotov, 2010a).

Definition: Factory-gate price indexes

p_i , from the market-clearing condition (2-4), directly measures how producer prices respond to trade-cost changes, and is the essential link for nesting gravity inside richer production structures (Lesson 3 and beyond).

Understanding Quiz 4 — Lesson 4

Q4.1 The inward multilateral resistance P_j can be interpreted as a:

- (A) supplier access index summarizing the trade-cost incidence facing j 's consumers
- (B) measure of j 's labour force
- (C) tariff schedule
- (D) random error term

Q4.2 P_j may only loosely track an observed consumer price index because:

- (A) it is measured in a different currency
- (B) it also captures intermediate-goods incidence, is production-weighted, may absorb home bias, and relies on the CES functional form
- (C) CPI data are never available
- (D) it only applies to services

5 Lesson 5: Indexes independent of normalisation — CTB, terms of trade, and the ACR welfare formula

Definition: Constructed Trade Bias (CTB)

Agnosteva et al. (2014) define

$$CTB_{ij} \equiv \frac{\hat{X}_{ij}}{Y_i E_j / Y} = \left(\frac{\hat{t}_{ij}}{\hat{\Pi}_i \hat{P}_j} \right)^{1-\sigma}, \quad (2-22)$$

the ratio of predicted to *frictionless* trade — i.e. the whole trade-cost term of the gravity decomposition (Chapter 1, Lesson 3), constructed *directly* from estimated coefficients and fixed effects.

Intuition: Four reasons CTB is the most flexible GE index

(i) It is **independent of σ** by construction, since it is built from the power-transformed objects $t_{ij}^{1-\sigma}$, $\Pi_i^{1-\sigma}$, $P_j^{1-\sigma}$ directly. (ii) As a fitted/conditional expectation it **inherits gravity's good fit**, smoothing out noisy, mismeasured bilateral flows. (iii) It **aggregates naturally**: the constructed home bias (CHB, Anderson and Yotov, 2010) is predicted-to-frictionless *internal* trade; the constructed foreign bias (CFB) and constructed domestic bias (CDB) indexes (Anderson et al., 2014) split international vs. domestic effects. (iv) It has a **data-only counterpart**, the trade bias index $TB_{ij} \equiv X_{ij}/(Y_i E_j/Y)$ (2-23), useful without even estimating a gravity model, at the cost of more measurement error.

Definition: Terms of trade

$$ToT_i \equiv \frac{\hat{p}_i}{\hat{P}_i}. \quad (2-24)$$

Under the endowment-economy interpretation, the numerator is a change in nominal income ($\hat{p}_i = p_i Q_i$'s growth factor) and the denominator a change in consumer prices, so ToT_i doubles as a real-income/welfare measure. This departs from the textbook “exporter price over importer price” ToT definition because $\hat{p}_i, \hat{P}_i$ include intra-national trade; excluding intra-national links from (2-2)–(2-3) recovers the conventional definition if desired.

Definition: Sufficient welfare statistics (Arkolakis, Costinot and Rodríguez-Clare, 2012)

A wide class of trade models delivers welfare gains as a function of exactly two “sufficient statistics”: the change in the domestic expenditure share $\lambda_{ii} \equiv X_{ii}/E_i$, and the trade elasticity. In the structural gravity model of this Guide,

$$WS_i \equiv \frac{W_i^{CFL}}{W_i^{BLN}} = \left(\frac{\lambda_{ii}^{CFL}}{\lambda_{ii}^{BLN}} \right)^{\frac{1}{1-\sigma}}, \quad W_i = \frac{E_i}{P_i}. \quad (2-25)$$

Intuition: Why the domestic share alone is (almost) enough

λ_{ii} summarizes, in a single observable number, everything about a country's *openness*: how much of its own consumption it still produces at home. ACR's insight is that, for a whole family of otherwise very different models (Armington, EK Ricardian, Melitz heterogeneous firms with a Pareto productivity distribution), the mapping from “how open you already are” to “how much you gain from getting more open” takes the identical CES-style power-function form, with only the trade elasticity differing across models. This is why (2-25) needs no data beyond baseline and counterfactual domestic trade shares plus σ (or θ).

Understanding Quiz 5 — Lesson 5

Q5.1 The Constructed Trade Bias (CTB) index is defined as:

- (A) actual trade divided by GDP
- (B) the ratio of predicted (fitted) trade to hypothetical frictionless trade

- (C) the elasticity of substitution
- (D) a tariff rate

Q5.2 The terms-of-trade index $ToT_i = \hat{p}_i / \hat{P}_i$ can be interpreted as a welfare measure because:

- (A) it is always equal to one
- (B) the numerator reflects nominal income and the denominator reflects consumer prices
- (C) it ignores trade costs
- (D) it only applies to non-member countries

Q5.3 The ACR sufficient-statistic welfare formula requires knowledge of:

- (A) the full bilateral trade cost matrix and all preference parameters
- (B) only the change in the domestic expenditure share and the trade elasticity
- (C) tariff revenue only
- (D) nothing beyond GDP

Part III

Counterfactual Analysis: the Standard Approach and GEPPML

Having defined the indexes, this Part turns to *computing* them: the generic five-step recipe for any general-equilibrium gravity counterfactual, and its most practical implementation — the GEPPML procedure of Anderson, Larch and Yotov (2015b), which needs nothing more exotic than repeated calls to a PPML gravity regression.

6 Lesson 6: The standard five-step approach

Derivation: Step 1 — Solve the baseline

Obtain baseline trade costs and elasticities via one of three routes:

- **Estimation:** PPML on (2-26) $X_{ij,t} = \exp[\pi_{i,t} + \chi_{j,t} + \mu_{ij} + \beta_1 RTA_{ij,t} + \beta_2 ES_{i,t} \times INTL_{ij} + \beta_3 MFN_{j,t} \times INTL_{ij}] \times \varepsilon_{ij,t}$, giving $\hat{t}_{ij,t}^{1-\sigma, BLN} = \exp[\hat{\mu}_{ij} + T_{ij,t}\hat{\beta}]$ (2-28). Advantages: accounts for measurement error, establishes causality, decomposes trade-cost components, extends out-of-sample (Box 3), and is externally checkable.
- **Calibration:** recover trade costs directly from data via ratio methods that cancel the country-specific structural terms — the *odds-ratio* method (Head and Ries, 2001; Novy, 2013), $\frac{t_{ij}t_{ji}}{t_{ii}t_{jj}} = \left(\frac{X_{ij}X_{ji}}{X_{ii}X_{jj}}\right)^{1/(1-\sigma)}$ (2-29); the *tetrads* method (Head et al., 2010) using two reference countries (2-30); the *three-countries* method (Caliendo and Parro, 2015) (2-31). Advantage: fits the data perfectly by construction. Disadvantage: cannot identify which *policy* produced a given cost change, and offers no specification test.
- **“Estibration”** (Anderson, Larch and Yotov, 2015b): a hybrid, $\hat{t}_{ij,t}^{1-\sigma, BLN} = \exp[\hat{\mu}_{ij} + T_{ij,t}\hat{\beta}] \times \hat{\varepsilon}_{ij,t}$ (2-32) — keeps the decomposability of estimation while matching the data exactly like calibration.

Then (optionally) recover any remaining structural parameters (e.g. CES weights α_i) so the model reproduces the data exactly (or skip this by solving the system *in changes*, à la Dekle,

Eaton and Kortum, 2007–2008), and finally construct all baseline indexes (Lesson 4–5).

Derivation: Step 2 — Define the counterfactual

Redefine the policy vector, e.g. flip an RTA dummy from 0 to 1 for a new pair, to get counterfactual costs $\hat{t}_{ij,t}^{1-\sigma,CFL} = \exp[\hat{\mu}_{ij} + T_{ij,t}^{CFL}\hat{\beta}]$ (2-33). The gap between (2-28)/(2-32) and (2-33) is the policy shock injected into the system.

Derivation: Step 3 — Solve the counterfactual model

Step 3.a (conditional GE): re-solve the multilateral resistance system (2-2)–(2-3) with the new $\hat{t}_{ij}^{CFL}$, holding Y_i, E_j fixed. **Step 3.b (full endowment GE):** re-solve the entire, now highly non-linear, $5 \times N$ -equation system (2-1)–(2-5) simultaneously for $\Pi_i, P_j, p_i, Y_i, E_i$. This can be done with a non-linear solver, or — as Lesson 7–8 show — with an iterative loop of ordinary PPML calls.

Derivation: Step 4 — Report; Step 5 — Confidence intervals

Step 4: report percentage changes, $\Delta\% \hat{I}_{i,t} = (\hat{I}_{i,t}^{CFL} - \hat{I}_{i,t}^{BLN})/\hat{I}_{i,t}^{BLN} \times 100$ (2-34), for any index of interest $\hat{I}$. **Step 5:** bootstrap. Anderson and Yotov (2016)’s recipe: (a) draw B (e.g. 200) bootstrapped gravity estimates; (b) construct B counterfactual cost matrices; (c) compute B sets of GE indexes; (d) form the centred bootstrap CI,

$$\left[2\hat{I}_{i,t} - \hat{I}_{i,t}^{BOOT(97.5\% \times B)} ; 2\hat{I}_{i,t} - \hat{I}_{i,t}^{BOOT(2.5\% \times B)} \right]. \quad (2-35)$$

Intuition: Why bother with confidence intervals for a GE simulation?

It is tempting to treat a GE counterfactual as a deterministic “black box.” But it is only as reliable as the Step 1 estimates that feed it, which themselves have sampling uncertainty. Reporting confidence intervals (i) tells a policy maker whether an eye-catching GE welfare number is actually statistically distinguishable from zero, and (ii) gives conservative and optimistic bounds for planning purposes — good practice that remains underused in the applied literature.

Understanding Quiz 6 — Lesson 6

Q6.1 The “calibration” approach to obtaining baseline trade costs has the advantage of fitting the data perfectly, but the disadvantage that it:

- (A) cannot identify which policy produced a given trade-cost change
- (B) requires panel data
- (C) cannot be used with PPML
- (D) ignores multilateral resistances

Q6.2 “Estimation” (Anderson, Larch and Yotov, 2015b) combines:

- (A) two different elasticities of substitution
- (B) the decomposability of estimation with the perfect data fit of calibration, by folding the error term into the trade-cost vector

- (C) OLS and Tobit
- (D) cross-section and panel data only

Q6.3 The “conditional general equilibrium” step of the standard approach solves:

- (A) the full $5 \times N$ system including prices and incomes
- (B) only the multilateral resistance system (2-2)–(2-3), holding output and expenditure fixed
- (C) nothing; it is skipped in practice
- (D) only equation (2-1)

7 Lesson 7: GEPPML — baseline and conditional general equilibrium via PPML fixed effects

Intuition: The key trick: PPML fixed effects *are* the structural terms

Because of PPML’s additive property (Arvis and Shepherd, 2013; Fally, 2015), the estimated exporter-time and importer-time fixed effects from a gravity regression map *exactly* onto the multilateral resistances:

$$\exp(\hat{\pi}_{i,t}) = \frac{Y_{i,t}}{\hat{\Pi}_{i,t}^{1-\sigma}} \times E_{R,t}, \quad \exp(\hat{\chi}_{j,t}) = \frac{E_{j,t}}{\hat{P}_{j,t}^{1-\sigma}} \times \frac{1}{E_{R,t}}. \quad (2-36, 2-37)$$

This means a full non-linear GE solver is *never strictly necessary*: repeatedly re-estimating a constrained PPML gravity model and reading off its fixed effects delivers the multilateral resistances directly.

Derivation: Step I — Baseline, GEPPML style

Estimate (2-26) by PPML, controlling for $\pi_{i,t}, \chi_{j,t}, \mu_{ij}$, with the reference country’s importer fixed effect normalized out ($P_{R,t} = 1$) and no constant term. Then:

$$\hat{\Pi}_{i,t}^{1-\sigma, BLN} = \frac{Y_{i,t}}{\exp(\hat{\pi}_{i,t})} \times E_{R,t} \quad \hat{P}_{j,t}^{1-\sigma, BLN} = \frac{E_{j,t}}{\exp(\hat{\chi}_{j,t})} \times \frac{1}{E_{R,t}} \quad (2-38, 2-39)$$

Unlike the standard approach, GEPPML’s “Step 1.b” (recovering α_i) is unnecessary: the iterative full-endowment procedure of Lesson 8 solves the system *in changes* (Dekle, Eaton and Kortum), so CES weights cancel automatically.

Intuition: Box 3 — Filling in missing trade-cost data

If a country pair has no non-zero trade for the entire sample, its pair fixed effect μ_{ij} cannot be estimated directly. Anderson and Yotov (2016)’s fix: (1) estimate $\hat{\mu}_{ij}$ for pairs with data; (2) regress those estimates on standard gravity covariates plus country fixed effects, $\exp(\hat{\mu}_{ij}) = \exp[\pi_i + \chi_j + \beta_1 \ln DIST_{ij} + \beta_2 CNTG_{ij} + \beta_3 LANG_{ij} + \beta_4 CLNY_{ij}] \times \varepsilon_{ij}$; (3) use the fitted values to fill the missing cells, producing a *complete* trade-cost matrix — essential for a counterfactual to capture the full general-equilibrium response, since omitted players bias everyone else’s multilateral resistances.

Derivation: Step II–III.a — Counterfactual and conditional GE

Redefine the policy vector $T_{ij,t}^{CFL}$ (Step II), then re-estimate the constrained PPML model

$$X_{ij,t} = \exp [\pi_{i,t}^{CFL} + \chi_{j,t}^{CFL} + \mu_{ij} + T_{ij,t}^{CFL} \hat{\beta}] \times \varepsilon_{ij,t}^{CFL} \quad (2-40)$$

holding $\hat{\beta}$ fixed at its baseline value ($\hat{\cdot}$ marks a constrained coefficient). The new fixed effects give the conditional GE multilateral resistances

$$\hat{\Pi}_{i,t}^{1-\sigma,CDL} = \frac{Y_{i,t}}{\exp(\hat{\pi}_{i,t}^{CFL})} \times E_{R,t} \quad \hat{P}_{j,t}^{1-\sigma,CDL} = \frac{E_{j,t}}{\exp(\hat{\chi}_{j,t}^{CFL})} \times \frac{1}{E_{R,t}}. \quad (2-41, 2-42)$$

Understanding Quiz 7 — Lesson 7

Q7.1 The GEPPML procedure recovers the multilateral resistances directly from:

- (A) a separate non-linear solver only
- (B) the estimated exporter-time and importer-time fixed effects of a PPML gravity regression
- (C) the RESET test statistic
- (D) bilateral tariff schedules

Q7.2 If a country pair has zero recorded trade throughout the sample, its bilateral trade cost can be recovered by:

- (A) leaving it undefined, biasing the whole counterfactual
- (B) regressing the estimable pair fixed effects on standard gravity covariates and country fixed effects, then predicting the missing values
- (C) assuming it equals zero
- (D) using OLS on log trade only

Q7.3 In the “conditional general equilibrium” PPML step, the coefficients $\hat{\beta}$ on the policy vector are:

- (A) re-estimated freely
- (B) held fixed at their baseline estimated values while the fixed effects re-adjust
- (C) set to zero
- (D) irrelevant

8 Lesson 8: GEPPML — the full endowment general equilibrium iterative loop

Intuition: Why an iterative loop, not one shot

The conditional GE step (Lesson 7) captures only the multilateral-resistance channel. To capture the *full endowment* GE channel of Lesson 2 — prices, income, and expenditure all adjusting — Anderson, Larch and Yotov (2015b) show that repeatedly cycling through (i) price update, (ii) income/trade update, (iii) PPML re-estimation converges to the exact solution of the full non-linear system, using nothing more than a loop of ordinary PPML calls.

Derivation: The four-stage loop

(i) Update factory-gate prices. Using the market-clearing condition (2-4) and the fixed-effect mapping (2-41),

$$\Delta p_{i,t}^{CFL} = \frac{p_{i,t}^{CFL}}{p_{i,t}} = \left[\frac{\exp(\hat{\pi}_{i,t}^{CFL})/E_{R,t}^{CFL}}{\exp(\hat{\pi}_{i,t})/E_{R,t}} \right]^{\frac{1}{1-\sigma}}. \quad (2-43)$$

(ii) Update income, expenditure, and trade. $Y_{i,t}^{CFL} = (p_{i,t}^{CFL}/p_{i,t})Y_{i,t}$, $E_{j,t}^{CFL} = (p_{j,t}^{CFL}/p_{j,t})E_{j,t}$, and, applying the gravity equation itself to translate these changes into a new trade matrix,

$$X_{ij,t}^{CFL} = \frac{\hat{t}_{ij,t}^{1-\sigma,CFL}}{\hat{t}_{ij,t}^{1-\sigma}} \times \frac{Y_{i,t}^{CFL} E_{j,t}^{CFL}}{Y_{i,t} E_{j,t}} \times \frac{\Pi_{i,t}^{1-\sigma}}{(\Pi_{i,t}^{1-\sigma})^{CFL}} \times \frac{P_{j,t}^{1-\sigma}}{(P_{j,t}^{1-\sigma})^{CFL}} \times X_{ij,t}. \quad (2-44)$$

This is only a *first-order* update (it uses the conditional, not yet fully updated, multilateral resistances). **(iii) Re-estimate.** Re-run PPML on the updated $X_{ij,t}^{CFL}$ from (2-44) to get a fresh set of fixed effects, hence fresh Π_i, P_j — which feed back into stage (i) for a new round of price updates. **(iv) Iterate to convergence.** Repeat (i)–(iii) until $\Delta p_{i,t}^{CFL} \rightarrow$ a fixed point for every country. At convergence, the full endowment GE indexes are

$$\hat{\Pi}_{i,t}^{1-\sigma,FULL} = \frac{Y_{i,t}^{FULL}}{\exp(\hat{\pi}_{i,t}^{FULL})} E_{R,t}^{FULL} \quad \hat{P}_{j,t}^{1-\sigma,FULL} = \frac{E_{j,t}^{FULL}}{\exp(\hat{\chi}_{j,t}^{FULL})} \frac{1}{E_{R,t}^{FULL}} \quad (2-45, 2-46)$$

$$p_{i,t}^{FULL} = p_{i,t}^{BLN} \left[\frac{\exp(\hat{\pi}_{i,t}^{FULL})/E_{R,t}^{FULL}}{\exp(\hat{\pi}_{i,t}^{BLN})/E_{R,t}^{BLN}} \right]^{\frac{1}{1-\sigma}} \quad Y_{i,t}^{FULL} = \frac{p_{i,t}^{FULL}}{p_{i,t}^{BLN}} Y_{i,t}^{BLN}, \quad E_{i,t}^{FULL} = \phi_i Y_{i,t}^{FULL}. \quad (2-47, 2-48-2-49)$$

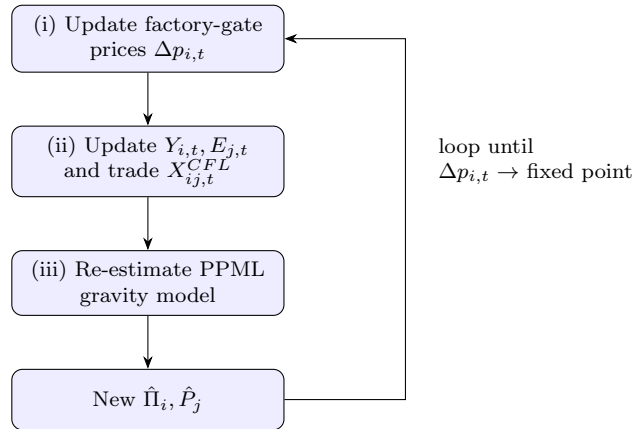


Figure 2: The GEPPML iterative loop for full endowment general equilibrium effects (after Figure 5 of the Guide).

Understanding Quiz 8 — Lesson 8

Q8.1 The GEPPML iterative loop converges when:

- (A) the trade elasticity reaches zero
- (B) the change in factory-gate prices from one iteration to the next goes to zero for every country
- (C) exactly 200 iterations have run
- (D) the pair fixed effects become negative

Q8.2 Equation (2-44), which updates trade in the loop, is described as only a “first-order” update because:

- (A) it ignores trade costs entirely
- (B) it uses the conditional (not yet fully updated) multilateral resistances, requiring further iteration to converge
- (C) it only applies to two countries
- (D) it assumes zero trade costs

Q8.3 The core practical appeal of GEPPML relative to a generic non-linear solver is that it:

- (A) requires no software at all
- (B) can be implemented with nothing more than repeated calls to a standard PPML routine available in most statistical packages
- (C) avoids the need for any baseline data
- (D) only works for two-country models

Part IV

Applications: Trade Without Borders, and NAFTA

Both applications use PPML/GEPPML on a 69-country manufacturing database (1986–2006), the same data underlying Chapter 1’s applications.

9 Lesson 9: Trade without borders

Derivation: Baseline estimation

Cross-section, 2006 data, reference country Germany:

$$X_{ij} = \exp \left[\pi_i + \chi_j \underbrace{-0.791}_{(0.050)} \ln DIST_{ij} + \underbrace{0.674}_{(0.107)} CNTG_{ij} \underbrace{-2.474}_{(0.119)} INTL_{ij} \right] \times \varepsilon_{ij} \quad (2-52)$$

Distance and contiguity estimates (−0.791, 0.674) are close to Head and Mayer’s (2014) meta-analysis medians (−0.89, 0.49). The international-border coefficient implies borders cut trade, on average, by $(\exp(-2.474) - 1) \times 100 \approx -91.6\%$ (s.e. 1.005, delta method) even after controlling for distance and contiguity — borders proxy for much more than geography.

Derivation: Counterfactual: $INTL_{ij} \rightarrow 0$ for every pair

Re-estimate with β_{INTL} constrained to zero (distance and contiguity coefficients held at their baseline values) to get the conditional GE response, then run the GEPPML loop (Lesson 8) for the full endowment GE response.

Intuition: Four findings from Figures 6–7

1. **Exports rise a lot, and more in full endowment than conditional GE:** the full-endowment response exceeds the conditional-GE response by 41 to 91 percentage points across countries, as size effects amplify the pure trade-cost effect.
2. **Bigger countries gain more in exports:** a strong positive correlation between (log) output and the export response to removing borders.
3. **Real GDP effects are large and, on average, comparable to half the gains found by Costinot and Rodríguez-Clare (2014) moving all the way to autarky** — suggesting the world has captured, at best, roughly half of the potential gains from full economic integration.
4. **But small/less-developed economies gain *more* in real GDP**, even though large economies export more: the reconciliation is that gains for small countries come mostly through the **consumer side** (lower import prices, i.e. falling P_j), whereas large countries' gains are concentrated on the **producer side** (higher factory-gate prices, p_i) — consistent with Anderson and Yotov's (2010b) finding for Canada that most of the gains from globalization there accrued to producers.

Understanding Quiz 9 — Lesson 9

Q9.1 In the “trade without borders” application, the estimated international-border coefficient implies borders reduce trade by approximately:

- (A) 9%
- (B) 50%
- (C) 91.6%
- (D) 2.5%

Q9.2 The full endowment general equilibrium export response to removing borders is:

- (A) always smaller than the conditional GE response
- (B) consistently larger than the conditional GE response, by 41 to 91 percentage points
- (C) exactly equal to the conditional GE response
- (D) unrelated to country size

Q9.3 Real GDP gains from removing all borders are found to be, on average:

- (A) larger for small/less-developed economies, concentrated on the consumer-price side
- (B) larger for small economies, concentrated on the producer-price side
- (C) identical across all countries
- (D) largest for the United States and Japan

10 Lesson 10: NAFTA in general equilibrium

Derivation: Results, Table 4 (selected countries)

Country	% Δ exports (partial)	% Δ exports (cond. GE)	% Δ exports (full GE)	% Δ real GDP (full GE)
CAN	55.05	34.99	37.46	3.40
MEX	56.65	41.64	43.51	3.81
USA	18.33	14.48	14.88	0.33
DEU (non-member)	0.00	-0.14	-0.17	-0.01
CHN (non-member)	0.00	-0.34	-0.38	-0.01
COL (non-member)	0.00	-1.80	-1.73	-0.03

Intuition: Reading the NAFTA results

- **Partial vs. conditional GE:** member-country exports increase far less in conditional GE (15–42%) than in partial equilibrium (19–57%), because trade diversion (Lesson 2) siphons off part of the gain — some of the new trade with partners comes at the expense of trade with (and within) non-members.
- **Conditional vs. full endowment GE:** full endowment export gains are *slightly larger* than conditional GE for members — producers there enjoy higher factory-gate prices, adding a size effect on top of the pure trade-cost effect — while the negative effect on many non-members is *mitigated*, because members' greater purchasing power (higher E_i) partly restores non-members' exports to them.
- **Welfare:** full endowment real GDP rises for all three NAFTA members (0.3 to 3.8%), essentially zero or very slightly negative for non-members — decomposed (columns 5–7 of Table 4) into falling inward multilateral resistances (consumer gains) and falling outward multilateral resistances/rising factory-gate prices (producer gains) for members, and the mirror image for many non-members.
- **Magnitude ordering:** non-member effects are two orders of magnitude smaller than member effects, illustrating the “first-order vs. second-order” distinction of Lesson 2 quantitatively, not just qualitatively.

Understanding Quiz 10 — Lesson 10

Q10.1 Member countries' export gains from NAFTA are smaller in conditional general equilibrium than in partial equilibrium because:

- (A) of measurement error only
- (B) trade diversion: part of the increase in trade with partners comes at the expense of trade with non-members and domestic sales
- (C) NAFTA reduced tariffs to zero on all goods
- (D) conditional GE ignores multilateral resistances

Q10.2 Non-member countries' export and real GDP responses to NAFTA are:

- (A) exactly zero in every scenario
- (B) small and typically negative, an order of magnitude smaller than the effects on members
- (C) larger than the effects on members
- (D) always positive

Q10.3 Full endowment general equilibrium export gains for NAFTA members are slightly

larger than conditional GE gains because:

- (A) of a coding convention
- (B) members' higher factory-gate prices and incomes add a size effect that reinforces the pure trade-cost effect
- (C) conditional GE overstates the true gains
- (D) non-members' trade costs fell too

Final Exam: Comprehensive Multiple-Choice Quiz

Understanding Quiz — Final Exam (all parts)

F1. The multilateral resistance system (2-2)–(2-3) can only be solved:

- (A) in unique absolute levels
- (B) up to a normalising scalar, requiring a reference-country normalisation
- (C) only with cross-section data
- (D) only if trade costs are symmetric

F2. A bilateral RTA between i and j affects non-member countries mainly through:

- (A) a direct cut in their own trade costs
- (B) changes in their own multilateral resistances, triggered by the liberalizing countries' improved market access
- (C) changes in their tariff schedules
- (D) nothing; non-members are unaffected

F3. In the “conditional general equilibrium” scenario, which of the following remains fixed?

- (A) bilateral trade costs
- (B) output and expenditure
- (C) the multilateral resistances
- (D) factory-gate prices

F4. In the “full endowment general equilibrium” scenario, relative to conditional GE, the additional channel activated is:

- (A) endogenous factory-gate prices, output, and expenditure
- (B) endogenous elasticities of substitution
- (C) endogenous population growth
- (D) endogenous physical endowments Q_i

F5. Nesting gravity inside a dynamic capital-accumulation model, setting $\delta = 0$:

- (A) breaks the gravity system
- (B) recovers the static endowment-economy gravity system exactly
- (C) makes trade costs irrelevant
- (D) eliminates the multilateral resistances

F6. The Constructed Trade Bias (CTB) index is independent of the elasticity of substitution σ because it is built from:

- (A) raw trade values only
- (B) the power-transformed objects $t_{ij}^{1-\sigma}$, $\Pi_i^{1-\sigma}$, $P_j^{1-\sigma}$ directly

- (C) tariff data
 - (D) GDP per capita
- F7.** The ACR sufficient-statistic welfare formula relates welfare gains to changes in:
- (A) the domestic expenditure share and the trade elasticity
 - (B) the exchange rate
 - (C) population growth
 - (D) government spending
- F8.** The GEPPML procedure recovers the multilateral resistances from:
- (A) a stand-alone non-linear solver only
 - (B) the estimated exporter-time and importer-time fixed effects of a PPML gravity regression
 - (C) survey data
 - (D) bilateral exchange rates
- F9.** The GEPPML iterative loop for full endowment GE effects converges when:
- (A) exactly one iteration has run
 - (B) the change in factory-gate prices across iterations goes to zero
 - (C) the pair fixed effects are dropped
 - (D) σ reaches its maximum value
- F10.** Missing/zero bilateral trade data over a whole sample period can be handled in the GEPPML framework by:
- (A) ignoring the pair permanently
 - (B) regressing estimable pair fixed effects on gravity covariates and country fixed effects, then predicting the missing values
 - (C) setting the missing trade cost to zero
 - (D) excluding the country entirely from all counterfactuals
- F11.** In the “trade without borders” application, real GDP gains from removing all international borders are found to be:
- (A) larger for small/less-developed economies, working mainly through lower consumer prices
 - (B) identical across all countries
 - (C) larger only for the United States and Japan
 - (D) negative for every country
- F12.** NAFTA member countries’ full endowment general equilibrium export gains are slightly larger than their conditional GE gains because:
- (A) of a size effect: higher factory-gate prices and income reinforce the pure trade-cost effect
 - (B) non-member trade costs also fell
 - (C) of a calculation error in the Guide
 - (D) conditional GE always overstates true effects
- F13.** Non-member countries’ trade and welfare responses to NAFTA are, in general equilibrium:
- (A) as large as member countries’ responses
 - (B) small, typically slightly negative, and an order of magnitude smaller than member responses
 - (C) always exactly zero
 - (D) always positive and large

F14. The core reason PPML fixed effects can substitute for a generic non-linear GE solver is:

- (A) PPML fixed effects are unrelated to the multilateral resistances
- (B) PPML's additive property makes its fixed effects map exactly onto the structural multilateral resistance terms
- (C) non-linear solvers are always faster
- (D) PPML cannot estimate fixed effects

F15. The dynamic gravity extension shows that a trade-cost shock in one country can affect capital accumulation everywhere because:

- (A) capital accumulation depends on factory-gate prices and the inward multilateral resistance, both of which are general-equilibrium objects that respond to shocks anywhere in the world
- (B) capital is perfectly mobile across countries
- (C) trade costs directly enter the capital accumulation equation with no other channel
- (D) dynamic considerations are irrelevant to gravity

A Answer key

Lesson quizzes

Lesson 1

Q1.1: B. Remote countries have fewer good alternatives, so bilateral links matter more (Krugman's Mars intuition). **Q1.2:** B. System (2-2)–(2-3) is homogeneous of degree zero in $\{\Pi_i, P_j\}$, requiring a normalisation. **Q1.3:** B. Π_i aggregates the trade-cost incidence facing producers in i .

Lesson 2

Q2.1: B. Conditional GE freezes output and expenditure while letting multilateral resistances adjust. **Q2.2:** B. Each $\Delta\Pi_i, \Delta P_j$ is a size-weighted average across all bilateral costs, so it is always smaller than Δt_{ij} itself. **Q2.3:** B. A fall in Π_i raises p_i via (2-4), raising $Y_i = p_i Q_i$. **Q2.4:** B. Partial equilibrium (narrowest) $\rightarrow$ conditional GE $\rightarrow$ full endowment GE (broadest).

Lesson 3

Q3.1: B. $\delta = 0$ shuts off capital accumulation, collapsing the system to the static one. **Q3.2:** B. Higher $p_{i,t}$ raises the marginal product of capital in the accumulation equation. **Q3.3:** B. $P_{i,t}$ prices both consumption and investment goods, so it is doubly relevant to saving decisions.

Lesson 4

Q4.1: A. P_j is a CES-based supplier access index for consumers. **Q4.2:** B. Multiple conceptual gaps separate P_j from a measured CPI.

Lesson 5

Q5.1: B. $CTB_{ij} = \hat{X}_{ij}/(Y_i E_j/Y)$, predicted over frictionless trade. **Q5.2: B.** $\hat{p}_i$ tracks nominal income, $\hat{P}_i$ tracks consumer prices. **Q5.3: B.** ACR's sufficient statistics: $\Delta\lambda_{ii}$ and the trade elasticity.

Lesson 6

Q6.1: A. Calibration cannot attribute a cost change to a specific policy or test the model. **Q6.2: B.** Estimation folds the residual into the trade-cost vector, matching data exactly while retaining decomposability. **Q6.3: B.** Conditional GE only re-solves (2-2)–(2-3), holding Y_i, E_j fixed.

Lesson 7

Q7.1: B. Equations (2-36)–(2-37) map PPML fixed effects onto Π_i, P_j . **Q7.2: B.** The two-stage Anderson-Yotov (2016) procedure fills missing pairs. **Q7.3: B.** $\hat{\beta}$ is held at its baseline (constrained) value in the conditional GE re-estimation.

Lesson 8

Q8.1: B. Convergence is defined by $\Delta p_{i,t} \rightarrow$ a fixed point. **Q8.2: B.** Equation (2-44) uses only the conditional multilateral resistances at that iteration. **Q8.3: B.** GEPPML only needs a loop of standard PPML calls, not a custom non-linear solver.

Lesson 9

Q9.1: C. $(\exp(-2.474) - 1) \times 100 \approx -91.6\%$. **Q9.2: B.** Full endowment effects exceed conditional GE by 41–91 percentage points. **Q9.3: A.** Small/less-developed economies gain more, mainly via lower consumer prices.

Lesson 10

Q10.1: B. Trade diversion from non-members and domestic sales shrinks the conditional GE gain relative to partial equilibrium. **Q10.2: B.** Non-member effects are small, mostly negative, and an order of magnitude smaller than members'. **Q10.3: B.** Higher member factory-gate prices/income add a reinforcing size effect.

Final exam

Final Exam key

F1: B. Homogeneity of degree zero requires a normalisation.
F2: B. Non-members are affected via their own multilateral resistances' response to members' improved access.
F3: B. Conditional GE fixes output and expenditure.
F4: A. Full endowment GE endogenizes prices, output and expenditure.
F5: B. $\delta = 0$ recovers the static system exactly.

- F6: B.** CTB is built from power-transformed structural terms, cancelling σ .
- F7: A.** Domestic expenditure share and the trade elasticity are ACR's sufficient statistics.
- F8: B.** GEPPML reads Π_i, P_j off PPML fixed effects.
- F9: B.** Convergence: $\Delta p_{i,t} \rightarrow$ fixed point across iterations.
- F10: B.** The two-stage pair fixed-effect regression fills missing bilateral costs.
- F11: A.** Small economies gain more, mainly through the consumer-price channel.
- F12: A.** Higher member prices/income reinforce the trade-cost effect in full endowment GE.
- F13: B.** Non-member responses are small, usually negative, and far smaller than members'.
- F14: B.** PPML's additive property equates its fixed effects with the structural terms.
- F15: A.** Capital accumulation depends on general-equilibrium prices and multilateral resistances, transmitting shocks worldwide.